

## LECTURE 14 (OCTOBER 29TH)

**Recall.** We have previously observed that for connected terms, the S-matrix elements could be written as the following using the LSZ reduction:

$$S_{fi} = (2\pi)^4 \delta^{(4)} \left( \sum_{i=1}^n p_i - \sum_{j=1}^m q_j \right) \prod_{i=1}^n \frac{1}{\sqrt{(2\pi)^3 \cdot 2E_{p_i}}} \prod_{j=1}^m \frac{1}{\sqrt{(2\pi)^3 \cdot 2E_{q_j}}} \cdot i\mathcal{M}_{fi}$$

**Definition.** (Scattering cross sections,  $n = 2 \rightarrow m$ ) We define the scattering cross section as

$$\begin{aligned} \sigma &= \frac{(\text{transition probability per unit time \& volume})}{(\text{flux of incident particles}) \times (\text{number density of target particles})} \\ &= \frac{1}{|\vec{J}_{\text{inc}}| \cdot \frac{1}{V}} \times \frac{|S_{fi}|^2 \times \left( \frac{1}{V/(2\pi)^3} \right)^{2+m}}{T \times V} \end{aligned}$$

Note that the normalisation of the momentum eigenstates were given as  $\langle \vec{p} | \vec{p} \rangle = \delta^{(3)}(\mathbf{0})$ . Then,

$$V = \int d^3\vec{x} = (2\pi)^3 \int \frac{d^3\vec{x}}{(2\pi)^3} e^{i\mathbf{0} \cdot \vec{x}} = (2\pi)^3 \cdot \delta^{(3)}(\mathbf{0})$$

The normalisation of the vectors thus become

$$\frac{1}{\sqrt{V/(2\pi)^3}} |\vec{p}\rangle$$

which justifies the substitution above. Then,

$$\begin{aligned} \sigma &= \frac{1}{|\vec{J}_{\text{inc}}| \cdot T} \frac{\left( (2\pi)^4 \cdot \delta^{(4)}(p_1 + p_2 - \sum_j p_j) \right)^2}{(V(2\pi)^3)^{2+m}} |\mathcal{M}_{fi}|^2 \\ &\quad \times \frac{1}{(2\pi)^3 \cdot 2E_{p_1}} \cdot \frac{1}{(2\pi)^3 \cdot 2E_{p_2}} \cdot \prod_{j=1}^m \frac{1}{(2\pi)^3 \cdot 2E_{q_j}} \end{aligned}$$

Note that  $(2\pi)^4 \delta^{(4)}(\mathbf{0}) = V \cdot T$  and that  $\vec{q}_j \sim \vec{q}_j + d\vec{q}_j$

$$\sigma = \frac{1}{|\vec{J}_{\text{inc}}| \cdot V} (2\pi)^4 \cdot \delta^{(4)}(p_1 + p_2 - \sum_{j=1}^m q_j) |\mathcal{M}_{fi}|^2 \frac{2E_{p_1} \cdot 2E_{p_2}}{\prod_{j=1}^m 2E_{q_j}} \frac{1}{V}$$

The differential scattering cross section for final states within the infinitesimal element  $d^3\vec{q}_j$  is then

$$d\sigma = \sigma \times \prod_{j=1}^m \frac{V d^3\vec{q}_j}{(2\pi)^3}$$

noting that the number of states within  $\vec{q}_j \sim \vec{q}_j + d^3\vec{q}_j$  is equal to

$$\frac{V d^3\vec{q}}{(2\pi)^3} \leftrightarrow \frac{1}{(2\pi)^3/V}$$

which is the density of momentum states. We conclude:

$$d\sigma = \frac{1}{|\vec{J}_{\text{inc}}| \cdot V} \frac{1}{2E_{p_1} \cdot 2E_{p_2}} \cdot (2\pi)^4 \delta^{(4)}(p_1 + p_2 - \sum_j q_j) |\mathcal{M}_{fi}|^2 \cdot \prod_{j=1}^m \frac{d^3\vec{q}_j}{(2\pi)^3 \cdot 2E_{q_j}}$$

replacing  $|\vec{J}_{\text{inc}}| \cdot V = 1/|\vec{v}_1 - \vec{v}_2|$

$$d\sigma = \frac{1}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} (2\pi)^4 \delta^{(4)}(p_1 + p_2 - \sum_j p_j) |\mathcal{M}_{fi}|^2 \prod_{j=1}^m \frac{d^3\vec{q}_j}{(2\pi)^3 \cdot 2E_{q_j}}$$

written in a Lorentz covariant manner.

**Example.** (Special case with  $m = 2, 2 \rightarrow 2$  scattering) In the center of mass frame ( $\vec{p}_1 + \vec{p}_2 = 0 = \vec{q}_1 + \vec{q}_2$ ),

$$\begin{aligned} \sigma &= \int d\sigma = \int \frac{d^3\vec{q}_1}{(2\pi)^3 \cdot 2E_{q_1}^{m_1} \cdot 2E_{q_2}^{m_2}} \frac{2\pi \cdot \delta(E_{p_1}^{M_1} + E_{p_2}^{M_2} - E_{q_1}^{m_1} - E_{-q_1}^{m_2})}{|\vec{v}_1 - \vec{v}_2| \cdot 2E_{p_1}^{M_1} \cdot 2E_{p_2}^{M_2}} \cdot |\mathcal{M}_{fi}|^2 \\ &= \int \frac{q_1^2 dq_1 d\Omega}{(2\pi)^2 \cdot 2E_{q_1}^{m_1} \cdot 2E_{-q_1}^{m_2}} \cdot \frac{\delta(E_{CM} - \sqrt{q_1^2 + m_1^2} - \sqrt{q_1^2 + m_2^2})}{|\vec{v}_1 - \vec{v}_2| \cdot 2E_{p_1}^{M_1} \cdot 2E_{p_2}^{M_2}} |\mathcal{M}_{fi}|^2 \\ &= \int d\Omega \frac{1}{|\vec{v}_1 - \vec{v}_2| \cdot 2E_{p_1}^{M_1} \cdot 2E_{p_2}^{M_2}} \cdot \int dq_1 \cdot \frac{q_1^2}{16\pi^2 \cdot E_{q_1}^{m_1} \cdot E_{-q_1}^{m_2}} \frac{\delta(q_1 - q_1^{CM})}{\frac{q_1}{E_{q_1}^{m_1}} + \frac{q_1}{E_{-q_1}^{m_2}}} \cdot |\mathcal{M}_{fi}|^2 \\ &= \int d\Omega \frac{1}{|\vec{v}_1 - \vec{v}_2| 2E_{p_1}^{M_1} \cdot 2E_{p_2}^{M_2}} \cdot \frac{q_1^{CM}}{16\pi^2 \cdot E_{CM}} \cdot |\mathcal{M}_{fi}|^2 \end{aligned}$$

we thus have

$$\begin{aligned} \left(\frac{d\sigma}{d\Omega}\right)_{CM} &= \frac{1}{|\vec{v}_1 - \vec{v}_2| \cdot 2E_{p_1}^{M_1} \cdot 2E_{p_2}^{M_2}} \cdot \frac{q_1^{CM}}{16\pi^2 \cdot E_{CM}} |\mathcal{M}_{fi}|^2 \\ &= \frac{q_1^{CM}}{64\pi^2 \cdot p_1^{CM} E_{CM}^2} \cdot |\mathcal{M}_{fi}|^2 \end{aligned}$$

where  $\vec{v}_i = \vec{p}_i/E_{p_i}^{M_i}$ . We notice that in the massless limit,

$$\frac{1}{64\pi^2 \cdot E_{CM}^2} \cdot |\mathcal{M}_{fi}|^2$$

**Definition.** (Decay rates,  $n = 1 \rightarrow m$ ) The decay rate is defined as the probability per unit time that the particle will decay which is equal to one over the life-time over the

particle.

$$\begin{aligned}\Gamma &= \frac{|S_{fi}|^2 \left( \frac{1}{V/(2\pi)^3} \right)^{1+m}}{T} \\ &= (2\pi)^4 \delta^{(4)}(p_1 - \sum_{j=1}^m q_j) \cdot |\mathcal{M}_{fi}|^2 \frac{1}{2E_{p_1}} \cdot \prod_{j=1}^m \frac{1}{2E_{q_j} \cdot V}\end{aligned}$$

The differential decay rate is given by

$$d\Gamma = \Gamma \times \prod_{j=1}^m \frac{V d^3 \vec{q}}{(2\pi)^3}$$

**Remark.** (Correlation functions & path integrals) The grand question from all the above is – how do we compute

$$\begin{aligned}G^{(N)}(x_1, \dots, x_N) &= \langle 0 | T \hat{\phi}(x_1) \dots \hat{\phi}(x_N) | 0 \rangle \\ \tilde{G}^{(N)}(p_1, \dots, p_N) &= \int \prod_i d^4 x_i e^{-ip_i \cdot x_i} G^{(N)}(x_1, \dots, x_N)\end{aligned}$$

The answer is, path integrals!

**Remark.** (Path integrals in quantum mechanics) We consider a 1-dimensional quantum system with  $\hat{H}(\hat{q}, \hat{p}) = \hat{p}^2/2m + V(\hat{q})$ . Consider the following basis:  $\{|q; t\rangle_H\}$ .

$$|q; t\rangle_H = e^{-i\hat{H}(0-t)}|q\rangle$$

which is a state localised at  $q$  at time  $t$  in the Heisenberg picture. The transition amplitude is

$${}_H \langle q_f; t_f | q_i; t_i \rangle_H$$

The natural question is, what is the transition amplitude in terms of a path integral?

**Theorem.** (Phase space path integral) We split the time interval  $t_f - t_i$  into  $N+1$  segments

$$\langle q_f; t_f | q_i; t_i \rangle = \langle q_f | e^{-i\hat{H}(t_f - t_N)} e^{-i\hat{H}(t_N - t_{N-1})} \dots e^{-i\hat{H}(t_1 - t_i)} | q_i \rangle$$

placing the completeness relation between all of these,

$$\begin{aligned}\langle q_f; t_f | q_i; t_i \rangle &= \int dq_1 \dots dq_N \langle q_f | e^{-i\hat{H}(t_f - t_N)} | q_N \rangle \langle q_N | \dots | q_1 \rangle \langle q_1 | e^{-i\hat{H}(t_1 - t_i)} | q_i \rangle \\ &= \int dq_1 \dots dq_N \langle q_f; t_f | q_N; t_N \rangle \dots \langle q_1; t_1 | q_i; t_i \rangle\end{aligned}$$

where  $t_n = t_i + n \cdot \Delta t$  ( $n = 0, \dots, N+1$ ) and

$$\delta t = \frac{t_f - t_i}{N+1}$$

## LECTURE 15 (NOVEMBER 3RD)

**Recall.** We have previously seen the Hamiltonian  $\hat{H} = \hat{p}^2/2m + V(\hat{q})$  and split time intervals

$$t_n + t_i + n \cdot \frac{t_f - t_i}{N + 1}$$

where  $(t_f - t_i)/(N + 1)$  is  $\Delta t$ . We can then calculate

$${}_H \langle q_f; t_f | q_i; t_i \rangle_H = \int dq_1 \dots dq_N \langle q_f; t_f | q_N; t_N \rangle \dots \langle q_1; t_1 | q_i; t_i \rangle$$

**Lemma.** Let's rewrite each piece.

$$\begin{aligned} \langle q_{n+1}; t_{n+1} | q_n; t_n \rangle &= \int dp_n \langle q_{n+1} | e^{-i\hat{H}(\hat{q}, \hat{p}) \cdot \Delta t} | p_n \rangle \langle p_n | q_n \rangle \\ &= \int dp_n e^{-iH(q_{n+1}, p_n) \Delta t} \frac{1}{2\pi} e^{ip_n(q_{n+1} - q_n)} \end{aligned}$$

as  $\langle q_n | q_n \rangle = \exp(-ip_n q_n)/\sqrt{2\pi}$ . Note that the exponential only pulls out because of other orderings (e.g., Weyl). Taking the limit to  $N \rightarrow \infty$ ,  $(\Delta t \rightarrow 0)$ ,

$$= \int \frac{dp_n}{2\pi} e^{-iH(q_n, p_n) \Delta t} e^{ip_n \dot{q}_n \Delta t} (1 + O(\Delta t^2))$$

where  $\dot{q}_n = (q_{n+1} - q_n)/(t_{n+1} - t_n)$ . The resulting transition amplitude is

$$\begin{aligned} \langle q_f; t_f | q_i; t_i \rangle &= \lim_{N \rightarrow \infty} \int \frac{dp_0}{2\pi} \prod_{i=1}^N \frac{dp_i}{2\pi} \exp \left[ i \sum_{n=1}^N \left( p_n \dot{q}_n \Delta t - iH(q_n, p_n) \Delta t \right) \right] \\ &= \int_{q(t_i)=q_i}^{q(t_f)=q_f} [dq][dp] \exp \left[ i \int_{t_i}^{t_f} dt \left( p(t) \dot{q}(t) - H(q(t), p(t)) \right) \right] \end{aligned}$$

The expression in the exponential is called the 1st order Lagrangian,

$$L^{(1st)}(q, \dot{q}, p) = p(t) \dot{q}(t) - H(q(t), p(t))$$

Also, the path integral is called the phase space path integral with the measure

$$[dq][dp] = \lim_{N \rightarrow \infty} \frac{dp_0}{2\pi} \prod_{n=1}^{\infty} \frac{dq_n dp_n}{2\pi}$$

**Theorem.** (Configuration space path integral)

$$\begin{aligned} \int \frac{dp_n}{2\pi} \exp \left[ i \left( p_n \dot{q}_n - \frac{p_n^2}{2\pi} \right) \Delta t \right] &= \int \frac{dp_n}{2\pi} \exp \left[ -\frac{i\Delta t}{2m} (p_n - m\dot{q}_n)^2 + i\Delta t \frac{m\dot{q}_n^2}{2} \right] \\ &= \sqrt{\frac{m}{2\pi i \Delta t}} e^{i\Delta t \cdot m\dot{q}^2/2} \end{aligned}$$

using the key result  $\int dx e^{-ax^2} = \sqrt{\pi/a}$ . The resulting transition amplitude is

$$\begin{aligned}\langle q_f; t_f | q_i; t_i \rangle &= \lim_{N \rightarrow \infty} \int \left( \frac{m}{2\pi i \cdot \Delta t} \right)^{(N+1)/2} \prod_{n=1}^N dq_n \exp \left[ i \sum_{n=0}^N \left( \frac{1}{2} m \dot{q}_n^2 - V(q_n) \right) \cdot \Delta t \right] \\ &= \int_{q(t_i)=q_i}^{q(t_f)=q_f} \mathcal{D}q \exp \left[ i \int_{t_i}^{t_f} dt \left( \frac{1}{2} m \dot{q}(t)^2 - V(q(t)) \right) \right]\end{aligned}$$

The expression in the brackets is now the 2nd order Lagrangian

$$L^{(2\text{nd})}(q, \dot{q}) = \frac{1}{2} m \dot{q}^2 - V(q)$$

and the configuration space path integral measure is

$$\mathcal{D}q = \lim_{N \rightarrow \infty} \left( \frac{m}{2\pi i \Delta t} \right)^{(N+1)/2} \prod_{n=1}^N dq_n$$

**Definition.** (Correlation functions) The correlation functions can be obtained by applying functional derivatives to the transition amplitude! This is done through adding time-dependent external sources to the Hamiltonian.

$$\langle q_f; t_f | q_i, t_i \rangle_{f,h} = \int_{q(t_i)=q_i}^{q(t_f)=q_f} [dq][dp] \exp \left[ i \int_{t_i}^{t_f} dt \left( p\dot{q} - H + qf + ph \right) \right]$$

where  $f = f(t)$  and  $h = h(t)$ . Now consider a functional derivative on the transition amplitude with respect to the external source,  $f(t_*)$ .

$$\begin{aligned}\frac{\delta}{i\delta f(t_*)} \langle q_f; t_f | q_i; t_i \rangle_{f,h} |_{f=h=0} &= \int_{q(t_i)=q_i}^{q(t_f)=q_f} [dq][dp] q(t_*) \exp \left[ i \int_{t_i}^{t_f} dt L^{(1\text{st})} \right] \\ &= \int dq_* \int_{q(t_*)=q_*}^{q(t_f)=q_f} [dq][dp] \exp \left[ i \int_{t_*}^{t_f} dt L^{(1\text{st})} \right] \times q_* \int_{q(t_i)=q_i}^{q(t_*)} [dq][dp] \exp \left[ i \int_{t_i}^{t_*} dt L^{(1\text{st})} \right] \\ &= \int dq_* \langle q_{t_f}; t_f | q_*; t_* \rangle q_* \langle q_*; t_* | q_i; t_i \rangle \\ &= \int dq_* \langle q_f; t_f | e^{i\hat{H}t_*} \hat{q} | q_* \rangle \langle q_* | e^{-i\hat{H}t_*} | q_i; t_i \rangle =\end{aligned}$$

which is exactly the 1-point correlation function of a position operator. For a time-ordered  $N$ -point correlation function,

$$\frac{\delta}{i\delta f(t_i)} \cdots \frac{\delta}{i\delta h(t_N)} \langle q_f; t_f | q_i; t_i \rangle_{f,h} |_{f,h=0} = \langle q_f; t_f | T \hat{q}(t_1) \cdots \hat{p}(t_N) | q_i; t_i \rangle$$

**Theorem.** (Feynman's  $i\varepsilon$ -prescription)

$$\langle q_f; t_f | q_i; t_i \rangle_{f,h}^{(i\varepsilon)} = \int [dq][dp] \exp \left[ i \int_{t_i}^{t_f} dt \left( p\dot{q} - (1 - i\varepsilon)H(q, p) + qf + ph \right) \right]$$

As the setup, we localise external sources, meaning

$$f(t), h(t) \neq 0 \quad \text{only} \quad [t_a, t_b]$$

and we choose discrete energy eigenstates  $\{|n\rangle \mid n = 0, 1, 2, \dots\}$  and eigenvalues are  $\{E_n \mid n = 0, 1, 2, \dots\}$  and wavefunction  $\psi_n(q) = \langle q | n \rangle$ .

$$\begin{aligned} \langle q_f; \infty | q_i; -\infty \rangle_{f,h}^{(i\varepsilon)} &= \sum_{m,n=0}^{\infty} \langle q_f; \infty | m; t_b \rangle_{f,h}^{(i\varepsilon)} \langle m; t_b | n; t_b \rangle_{f,h}^{(i\varepsilon)} \langle n; t_a | q_i; -\infty \rangle_{f,h}^{(i\varepsilon)} \\ &= \sum_{m,n=0}^{\infty} \langle q_f | e^{-i(1-i\varepsilon)E_m(\infty-t_b)} | m \rangle \langle m; t_b | n; t_a \rangle_{f,h}^{(i\varepsilon)} \\ &\quad \times \langle n | e^{-i(1-i\varepsilon)E_n(t_a+\infty)} | q_i \rangle \\ &= \langle q_f | 0 \rangle \langle 0; t_b | 0; t_a \rangle_{f,h}^{(i\varepsilon)} \langle 0 | q_i \rangle \\ &= \psi_0(q_f) \psi_0(q_i)^* \langle 0; t_b | 0; t_a \rangle_{f,h}^{(i\varepsilon)} \end{aligned}$$

the front is independent of  $f$  and  $h$  which grants this the name the “vacumm persistence amplitude”.

$$\langle 0; t_b | 0; t_a \rangle_{f,h}^{(i\varepsilon)} = \mathcal{N} \int [dq][dp] \exp \left[ i \int dt \left( p\dot{q} - (1 - i\varepsilon)H + qf + ph \right) \right]$$

We comment that

- (i)  $\mathcal{N}$  is independent of  $f$  and  $h$ .
- (ii) The vacumm persistence amplitude is insensitive to the boundary position eigenstates.

**Corollary.** (Correlation function with respect to the vaccum state)

$$\frac{\delta}{i\delta f(t_i)} \cdots \frac{\delta}{i\delta h(t_N)} \langle 0; t_b | 0; t_a \rangle_{f,h}^{(i\varepsilon)} = \langle 0; t_b | T\hat{q}(t_i) \cdots \hat{p}(t_N) | 0; t_b \rangle^{(i\varepsilon)}$$

## LECTURE 16 (NOVEMBER 5TH)

**Remark.** (From quantum mechanics to quantum field theory) The map between the two pictures are as follows:

$$\begin{aligned}
 q(t) &\rightarrow \phi(x) \\
 p(t) = m\dot{q} &\rightarrow \pi(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi)} = \partial_0 \phi \\
 H = \frac{p^2}{2m} + V(q) &\rightarrow \mathcal{H}(\phi, \nabla \phi, \pi) = \frac{1}{2}\pi^2 + \frac{1}{2}|\nabla \phi|^2 + \frac{1}{2}m^2\phi^2 - \mathcal{L}_f \\
 L^{(2\text{nd})} = \frac{1}{2}m\dot{q}^2 - V(q) &\rightarrow \mathcal{L}(\phi, \partial_\mu \phi) = -\frac{1}{2}(\partial^\mu \phi)(\partial_\mu \phi) - \frac{1}{2}m^2\phi^2 + \mathcal{L}_f
 \end{aligned}$$

Recall that Feynman's  $i\epsilon$ -prescription in quantum field theory was

$$\mathcal{H} \rightarrow (1 - i\epsilon)\mathcal{H}$$

or equivalently  $m^2 \rightarrow m^2 - i\epsilon$ . The vacuum persistence amplitude in quantum field theory is thus

$$\langle 0; t_b | 0; t_a \rangle_{f,h} = \mathcal{N} \int [dq][dp] \exp \left[ i \int dt \left( p\dot{q} - (1 - i\epsilon)H + qf + ph \right) \right]$$

implying that

$$\begin{aligned}
 \langle 0|0 \rangle_J &= \int [d\phi][d\pi] \exp \left[ i \int d^4x \left( \mathcal{L}^{(1\text{st})}(\phi, \partial_\mu \phi, \pi) + \phi J \right) \right] \\
 &= \int \mathcal{D}\phi \exp \left[ i \int d^4x \left( \mathcal{L}^{(2\text{nd})}(\phi, \partial_\mu \phi) + \phi J \right) \right]
 \end{aligned}$$

The correlation functions in quantum field theory can then be written as

$$\frac{\delta}{i\delta J(x_1)} \cdots \frac{\delta}{i\delta J(x_N)} \langle 0|0 \rangle_{J=0} = \langle 0 | T \hat{\phi}(x_1) \cdots \hat{\phi}(x_N) | 0 \rangle$$

We make some comments:

- (i) Quantum field theory path integral measure includes a  $J$ -independent normalisation
- (ii)  $i\epsilon$ -prescription is implicitly assumed
- (iii)  $t_a \rightarrow -\infty$  and  $t_b \rightarrow \infty$  which is from now on omitted
- (iv) We assume  $\mathcal{L}^{(2\text{nd})} \rightarrow \mathcal{L}$  for notational convenience

The vacuum persistence amplitude is also called a “generating functional” for correlation functions. There are multiple types for this! For example,

- (i) (Generating functional)  $Z[J] = \langle 0|0 \rangle_J$  generates  $N$ -point functions

$$G^{(N)}(x_1, \dots, x_n) = \frac{\delta^N Z[J]}{i\delta J(x_1) \dots i\delta J(x_N)} = \langle T\hat{\phi}(x_1) \dots \hat{\phi}(x_N) \rangle$$

- (ii) (Normalised generating functional)

$$Z_n[J] = \frac{Z[J]}{Z[0]}$$

generates the normalised  $N$ -point function

$$G_n^{(N)}(x_1, \dots, x_N) = \frac{\delta^N Z_n[J]}{i\delta J(x_1) \dots i\delta J(x_N)}$$

Note that the normalisation kills the “vacuum bubble” diagrams.

- (iii) (Generating functional for “connected diagrams”)  $W[J] = \log Z[J]$  which generates connected  $N$ -point functions

$$G_c^{(N)}(x_1, \dots, x_N) = \frac{\delta^N W[J]}{i\delta J(x_1) \dots i\delta J(x_N)}$$

- (iv) (Generating functional for “1-particle irreducible” diagrams)

$$\Gamma[\phi_c] = W[J] - i \int d^4x \phi_c(x) J(x)$$

where

$$\phi_c(x) = \frac{\delta W[J]}{i\delta J(x)}$$

this generates the 1-PI  $N$ -point functions

$$G_{1PI}^{(N)}(x_1, \dots, x_N) = \frac{\delta^N \Gamma[\phi_c]}{i\delta\phi_c(x_1), \dots, i\delta\phi_c(x_N)}$$

here, 1 PI-diagrams are connected diagrams that cannot be split into two by cutting one leg.

**Example.** (Free spin-0, Klein-Gordan)

$$Z[J] = \int \mathcal{D}\phi \exp \left[ i \int d^4x \left( -\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}(m^2 - i\epsilon)\phi^2 + \phi J \right) \right]$$

- (i) The Fourier-transform takes

$$\tilde{\phi}(k) = \int d^4x e^{-ik \cdot x} \phi(x) \leftrightarrow \phi(x) = \int \frac{d^4x}{(2\pi)^4} e^{ik \cdot x} \tilde{\phi}(x)$$



which gives

$$Z[J] = \int \mathcal{D}\phi \exp \left[ \frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} b g i r - \tilde{\phi}(k)(k^2 + m^2 - i\epsilon)\tilde{\phi}(-k) + \tilde{J}(k)\tilde{\phi}(-k) + \tilde{J}(-k)\tilde{\phi}(k) \right]$$

(ii) Changing variables,

$$\begin{aligned} \chi(x) &= \phi(x) - \int \frac{d^4 k}{(2\pi)^4} e^{ikx} \cdot \frac{\tilde{J}(k)}{k^2 + m^2 - i\epsilon} \\ &= \int \frac{d^4 k}{(2\pi)^4} e^{ikx} \left( \tilde{\phi}(k) - \frac{\tilde{J}(k)}{k^2 + m^2 - i\epsilon} \right) \end{aligned}$$

and

$$\begin{aligned} Z[J] &= \int \mathcal{D}\chi \exp \left[ \frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} \left( -\tilde{\chi}(k)(k^2 + m^2 - i\epsilon)\tilde{\chi}(-k) + \frac{\tilde{J}(k)\tilde{J}(-k)}{k^2 + m^2 - i\epsilon} \right) \right] \\ &= \exp \left[ \frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} \cdot \frac{\tilde{J}(k)\tilde{J}(-k)}{k^2 + m^2 - i\epsilon} \right] \times \int \mathcal{D}\chi \exp \left[ \cdot \right] \end{aligned}$$

(iii) Setting  $Z[0] = 1$  for a free theory,

$$Z[J] = \exp \left[ \frac{i}{2} \int \frac{d^4 k}{(2\pi)^4} \cdot \frac{\tilde{J}(k)\tilde{J}(-k)}{k^2 + m^2 - i\epsilon} \right]$$

and taking the inverse transform

$$\tilde{J}(k) = \int d^4 x e^{-ikx} J(x)$$

$$Z[J] = \exp \left[ -\frac{1}{2} \int d^4 x d^4 x' J(x) \Delta_F(x - x') J(x') \right]$$

with

$$\Delta_F(x - x') = \int \frac{d^4 k}{(2\pi)^4} \frac{-i}{k^2 + m^2 - i\epsilon} e^{ik(x-x')}$$

Now, our correlation functions are, for odd number of functions,

$$G^{(2N+1)}(x_1, \dots, x_{2N+1}) = \frac{\delta^{2N+1} Z[J]}{i\delta J(x_1), \dots, i\delta J(x_{2N+1})} \Big|_{J=1} = 0$$

In the two particle case,

$$\begin{aligned} G^{(2)}(x_1, x_2) &= \langle T \hat{\phi}(x_1) \hat{\phi}(x_2) \rangle \\ &= \frac{\delta}{i\delta J(x_1)} \left( \frac{-1}{i} \int d^4 x J(x) \Delta_F(x - x_2) \right) Z[J] \Big|_{J=0} \\ &= \Delta_F(x_1 - x_2) \end{aligned}$$

For the even number case, we can conclude

$$G^{(2N)}(x_1, \dots, x_N) = \sum_{\sigma \in S_{2N}} G^{(2)}(x_{\sigma(1)} - x_{\sigma(2)}) \dots G^{(2)}(x_{\sigma(2N-1)} - x_{\sigma(2N)})$$

**Theorem.** (Wick theorem) The  $2N$ -point function in the above can be seen as the application of Wick's theorem. For example, consider the 4-point function

$$G^{(4)}(x_1, x_2, x_3, x_4) = G^{(2)}(x_1 - x_2)G^{(2)}(x_3 - x_4) + G^{(2)}(x_1 - x_3)G^{(2)}(x_2 - x_4) + G^{(2)}(x_1 - x_4)G^{(2)}(x_2 - x_3)$$

## LECTURE 17 (NOVEMBER 10TH)

**Remark.** We have seen how the generating function for Dirac theory was

$$Z[\bar{\eta}, \eta] = \exp \left[ - \int d^4x \int d^4y \bar{\eta}(x) S_F(x - y) \eta(y) \right]$$

where

$$S_F(x - y) = \int \frac{d^4k}{(2\pi)^4} \frac{-i}{\not{k} + m - i\epsilon} e^{-ik \cdot (x - y)}$$

**Theorem.** (Correlation functions from  $Z[\bar{\eta}, \eta]$ ) We see that

$$\begin{aligned} G_{\alpha;\beta}^{(1;1)}(x; y) &= \langle 0 | T \hat{\psi}_\alpha(x) \hat{\bar{\psi}}_\beta(y) | 0 \rangle \\ &= \frac{\delta}{i\delta\bar{\eta}^\alpha(x)} \cdot \frac{i\delta}{\delta\eta^\beta(y)} \\ &= S_F{}_{\alpha\beta}(x - y) \end{aligned}$$

For an arbitrary number of particles you have Wick's theorem for generalised fermions,

$$\begin{aligned} G_{\alpha_1, \dots, \alpha_N; \beta_1, \dots, \beta_N}^{(N;N)}(x_1, \dots, x_N; y_1, \dots, y_N) &= \langle T \hat{\psi}_{\alpha_1}(x_1) \dots \hat{\psi}_{\alpha_N}(x_N) \hat{\bar{\psi}}_{\beta_1}(y_1) \dots \hat{\bar{\psi}}_{\beta_N}(y_N) \rangle \\ &= (-1)^{N(N-1)/2} \sum \epsilon(\sigma) G_{\alpha_1, \beta_{\sigma(1)}}^{(1;1)}(x_1; y_{\sigma(1)}) \dots G_{\alpha_N, \beta_{\sigma(N)}}^{(1;1)}(x_N; y_{\sigma(N)}) \end{aligned}$$

**Example.** (4 point function)

$$G_{\alpha_1\alpha_2;\beta_1\beta_2}^{(2;2)}(x_1, x_2; y_1, y_2) = -G_{\alpha_1;\beta_1}^{(1;1)}(x_1; y_1)G_{\alpha_2;\beta_2}^{(1;1)}(x_2; y_2) + G_{\alpha_1;\beta_2}^{(1;1)}(x_1; y_2)G_{\alpha_2;\beta_1}^{(1;1)}(x_2; y_1)$$

**Definition.** (Path integrals in quantum field theory; vectors) The “naive” generating functional for the free Maxwell fields is

$$Z[J] = \int \mathcal{D}A_\mu \exp \left[ i \int d^4x \left( -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu \right) \right]$$

This is ill-defined due to gauge symmetry!

(i) Let's focus on the case with

$$\begin{cases} J^\mu \rightarrow 0 & \text{fast enough } x \rightarrow \infty \\ \partial_\mu J^\mu = 0 \end{cases}$$

- (ii) The integrand of the path integral is an invariant vector,  $A_\mu \rightarrow A_\mu^{(\lambda)} = A_\mu + \partial_\mu \lambda$
- (iii) The integration over the “gauge-equivalent” configurations of  $A_\mu$  leads to divergences. This requires gauge-fixing procedures.

**Proposition.** (Gauge-fixing procedure) (I) Let's temporarily write

$$Z[J] = \int \mathcal{D}A_\mu \left( \int \mathcal{D}\lambda |\det(\partial^\mu \partial_\mu)| \delta(\partial^\mu A_\mu^{(\lambda)} - \chi) \right) \exp \left[ i \int d^4x \left( -\frac{1}{4} F^2 + J \cdot A \right) \right]$$

Note that

$$1 = \int d(g(x)) \delta(g(x)) = \int dx |g'(x)| \delta(g(x))$$

for multiple variables,

$$\int d^n x \left| \det \frac{\partial g_i(x)}{\partial x_j} \right| \delta^n(g(x))$$

(II) Change of dummy field  $A_\mu \rightarrow A_\mu^{(-\lambda)}$ .

$$Z[J] = \int \mathcal{D}A_\mu^{(-\lambda)} \int \mathcal{D}\lambda |\det(\partial^2)| \delta(\partial^\mu A_\mu - \chi) \exp \left[ i \int d^4x \left( ibgr - \frac{1}{4} F^2 + J^\mu A_\mu \right) \right]$$

A few assumptions are that:

- (i) External currents are local and conserved
- (ii) Path-integral measure is not invariant under gauge symmetry. If the measure is not invariant, the symmetry is called anomalous. A gauge anomaly is pathological and a global anomaly is acceptable.

Step (III) comes from noticing that  $\mathcal{D}\lambda$  captures the divergence due to gauge symmetry. The well-defined generating functional is

$$Z_\chi[J] = \int \mathcal{D}A_\mu \delta(\partial^\mu A_\mu - \chi) \exp \left[ i \int d^4x \left( -\frac{1}{4} F^2 + J \cdot A \right) \right]$$

Let's make some comments:

- (i) The determinant is called the Faddeev-Popov (FP) determinant  $|\det(\partial^2)|$  only affects irrelevant normalisation only!
- (ii) Note that otherwise, we introduce FP ghosts to account for its contribution.

The last step (IV) comes from introducing the weight function

$$\begin{aligned} Z[J] &= \int \mathcal{D}\chi Z_\chi[J] \exp \left[ \frac{i}{2\xi} \int d^4x \chi^2 \right] \\ &= \int \mathcal{D}A_\mu \exp \left[ i \int d^4x \left( -\frac{1}{4} F^2 + \frac{1}{2\xi} (\partial^\mu A_\mu)^2 + J \cdot A \right) \right] \end{aligned}$$

where the second is the gauge-fixing term. Note that this does not affect the  $J$ -dependence of  $Z[J]$ ,  $Z[J]$  is not independent of  $\chi$ , and  $\xi = -1$  is called the Feynman gauge.

**Example.** (Free spin-1, Maxwell)

(i) Fourier transform gives

$$\hat{A}_\mu(k) = \int d^4x e^{-ikx} A_\mu(x)$$

this implies that

$$Z[J] = \int \mathcal{D}A_\mu \exp \left[ \frac{i}{2} \int \frac{d^4k}{(2\pi)^4} \left( -\tilde{A}_\mu(k) \left( (k^2 - i\epsilon) \eta^{\mu\nu} - \left( 1 + \frac{1}{\xi} \right) k^\mu k^\nu \right) \tilde{A}_\nu(-k) \right) \right]$$

(ii) Change of the path integral measure gives

$$D_{\mu\nu}(k) = \frac{-i(\eta_{\mu\nu} - (1 + \xi) \hat{k}_\mu \hat{k}_\nu)}{k^2 - i\epsilon}$$

if and only if

$$D_{\mu\nu}^{-1} = i \left( (k^2 - i\epsilon) \eta_{\mu\nu} - \left( 1 + \frac{1}{\xi} \right) k_\mu k_\nu \right)$$

Note that without gauge fixing ( $\xi = \infty$ ) implies that  $k^2 \eta^{\mu\nu} - k^\mu k^\nu$  is not invertible.

$$B_\mu(x) = A_\mu(x) - i \int \frac{d^4k}{(2\pi)^4} e^{ikx} D_{\mu\nu}(k) \tilde{J}^\nu(k)$$

implies that

$$Z[J] = \int \mathcal{D}B_\mu \exp \left[ -\frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \left( \tilde{B}_\mu(k) (D^{-1})^{\mu\nu}(k) \tilde{B}_\nu(-k) + \tilde{J}^\mu(k) D_{\mu\nu}(k) \tilde{J}^\nu(-k) \right) \right]$$

and

$$Z[J] = \exp \left[ \frac{-1}{2} \int \frac{d^4k}{(2\pi)^4} \tilde{J}^\mu(k) D_{\mu\nu}(k) \tilde{J}^\nu(-k) \right] \int \mathcal{D}B_\mu \exp \left[ -\frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \tilde{B}_\mu(k) (D^{-1})^{\mu\nu}_\mu(k) (P^{-1})^{\mu\nu}(k) \tilde{B}_\nu(k) \right]$$

(iii) And  $Z[0] = 1$

We therefore have, finally,

$$Z[J] = \exp \left[ -\frac{1}{2} \int d^4x d^4y J^\mu(x) D_{F\mu\nu}(x-y) J^\nu(y) \right]$$

where

$$D_{F\mu\nu}(x-y) = \int \frac{d^4k}{(2\pi)^4} \cdot \frac{-i(\eta_{\mu\nu} - (1+\xi)\hat{k}_\mu\hat{k}_\nu)}{k^2 - i\epsilon} e^{ik\cdot(x-y)}$$